

UNIT 3

PROGRAMMING

FUNDAMENTALS

9TH COMPUTER SCIENCE FEDERAL BOARD

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AGENDA

Basic Terminologies

HTML

Tags in HTML

Tags used in text

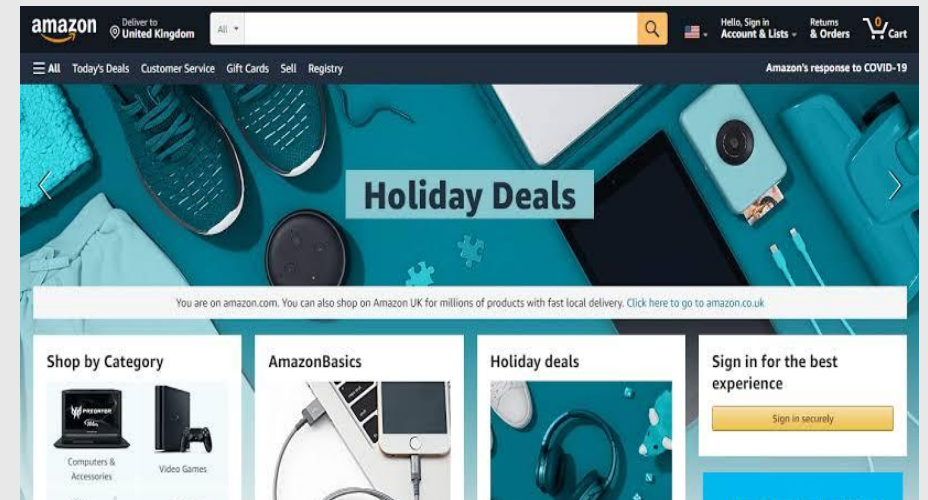
CSS

BASIC TERMINOLOGIES

Webpage – A single document or page on the internet. E.g. The "Contact Us" page on a website.



Website – A collection of many webpages linked together under one domain name. E.g. The entire Amazon site with all its pages.



BASIC TERMINOLOGIES CONT....

URL: It is the web address you type in a browser to visit a specific webpage e.g. <https://www.google.com>

Breakdown of a simple URL:

<https://www.example.com/home>

- <https://> – Protocol (rules for transferring data)
- [www.example.com](https://www.example.com/home) – Domain name (website's address)
- [/home](https://www.example.com/home) – Path (specific page or file)

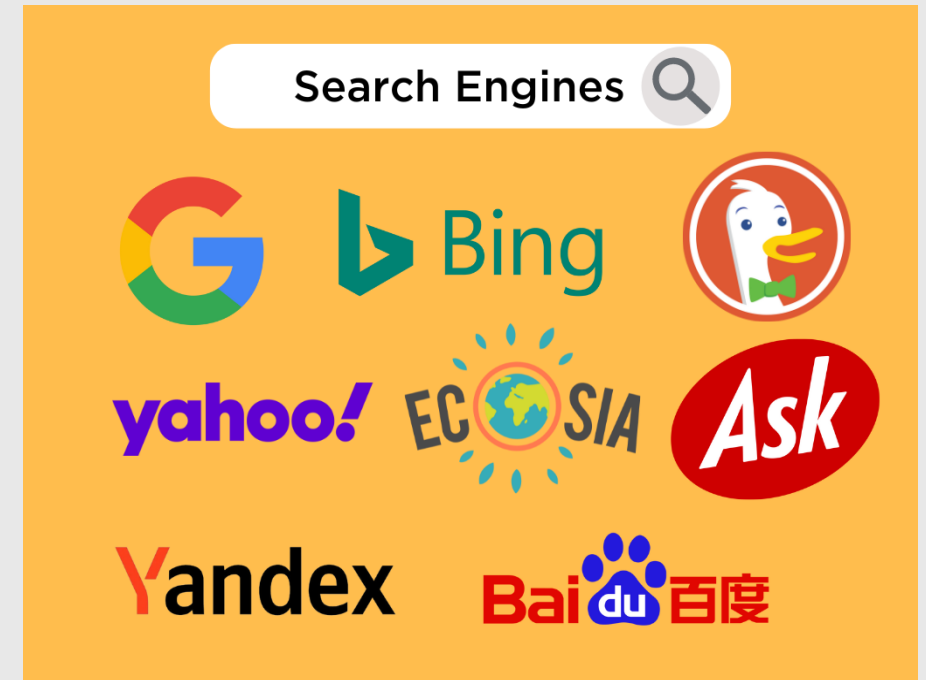


BASIC TERMINOLOGIES CONT....

Search Engine: It is a tool/website that helps you find information on the internet.

You type keywords or questions, and it shows you a list of relevant webpages.

Examples: Google, Bing, Yahoo, DuckDuckGo.



BASIC TERMINOLOGIES CONT....

Web Hosting Service: It is a service to stores your website's files (text, images, videos) on a server (a powerful computer) so that people can access it on the internet.

Without web hosting, your website cannot be viewed online.

Common web hosting providers: Hostinger, Bluehost, GoDaddy.



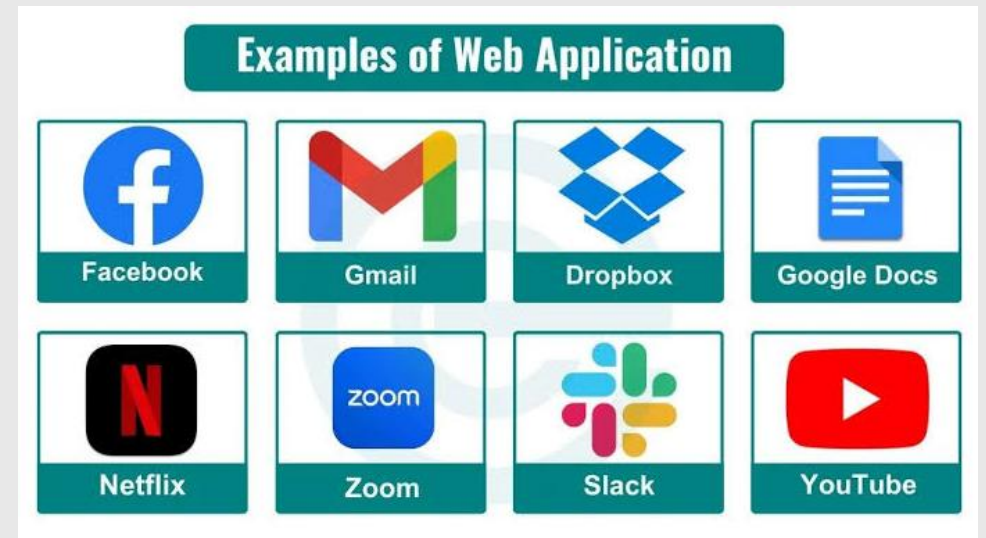
BASIC TERMINOLOGIES CONT....

Web Application: It is an interactive program that runs in browser and allows to do things (not just read information), like:

- *Send emails (Gmail)*
- *Edit documents (Google Docs)*
- *Shop online (Amazon)*
- *Book tickets (MakeMyTrip)*

Key difference from a normal webpage:

- **Webpage** = mostly read-only information (like a news article).
- **Web application** = lets you interact, enter data, and get results based on your actions.



BASIC TERMINOLOGIES CONT....

Static Website: A website where content stays the same for every user unless the developer manually changes the code.

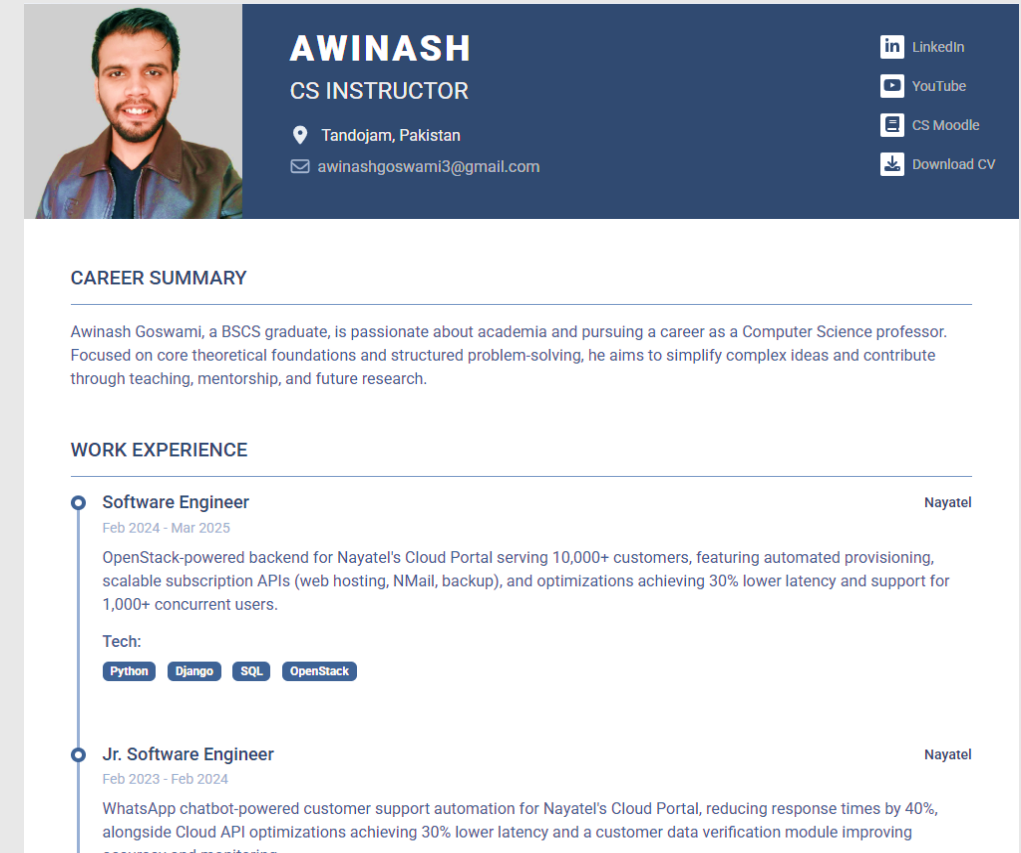
It is made using only HTML, CSS, and sometimes JavaScript (no databases or server-side processing).

Examples:

A personal portfolio page

A small business brochure site

A landing page with fixed information



BASIC TERMINOLOGIES CONT....

Dynamic Website: A website where content changes based on the user, time, or actions.

It uses server-side languages (like PHP, Python, or JavaScript) and databases to generate pages on the fly.

Content is not fixed – it is created in real-time when someone visits.

Examples:

- *Facebook (feed changes for each user)*
- *YouTube (recommendations based on your history)*
- *Amazon (product suggestions and personalized prices)*
- *Gmail (your inbox updates with new emails)*



BASIC TERMINOLOGIES CONT....

Front-End Development: It is the part of web development that deals with everything the user sees and interacts with directly in their browser. It focuses on the look, feel, and behavior of a website or web application.

Key Technologies:

- *HTML – Structure (headings, paragraphs, images)*
- *CSS – Styling (colors, fonts, layout)*
- *JavaScript – Interactivity (buttons, animations, forms)*

Examples of what front-end developers do:

- *Design menus, buttons, and forms*
- *Make a website look good on mobile and desktop*
- *Create smooth animations and pop-ups*
- *Ensure fast loading and easy navigation*



BASIC TERMINOLOGIES CONT....

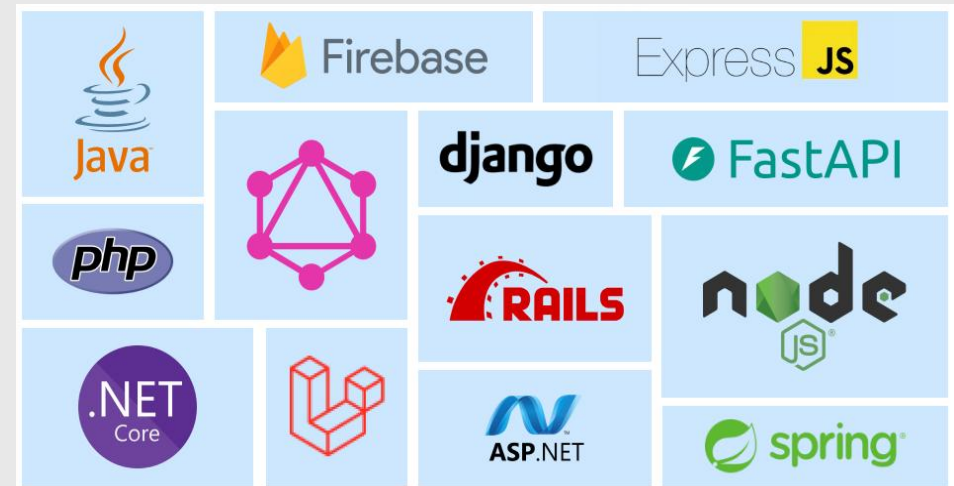
Back-End Development: It is the part of web development that deals with everything behind the scenes – the server, database, and application logic that users do not see. It makes sure the website works properly, stores data, and processes user requests.

Key Technologies:

- *Languages: Python, PHP, Java, Ruby, C#, JavaScript (Node.js)*
- *Databases: MySQL, PostgreSQL, MongoDB (to store user data)*
- *Servers: Apache, Nginx (to host and run the website)*

Examples of what back-end developers do:

- *Store user login details in a database*
- *Process online payments securely*
- *Send emails or notifications*
- *Fetch and send data to the front-end when a user clicks a button*



HTML

- Hypertext Markup Language (HTML) is used to define contents with help of tags for webpage.
- Tags are written with angle brackets '<' and '>'. For instance; <p> </p> tag used for paragraph and <h1> </h1> to define largest heading.
- The content is written between opening and closing tag such as <p> and </p> respectively.
- There are also some tag which don't need ending tags like
 <hr> for break line and horizontal line respectively.
- Actual Code example of tag and its content: <p> This is my first website </p> this line will display the in browser.
- You can write your html code in a notepad and run it in a browser.
- Other better software for the coding that help in minimizing your coding efforts are Visual Studio, NetBeans, etc.

TAGS IN HTML

- **html:** This is first tag of the HTML document, everything we see on browser lies inside the tag.
- **head:** This tag comes after <html> tag which is used to define metadata which is not for browser content but information about website like title and links like style.css and script.js etc.
- **title:** <title> tag is used define name of the website which we see on tab in a browser.
- **body:** <body> tag is one where we write our actual content like headings, paragraphs, links, buttons etc.
- **Tags for Headings:** There are six headings in html document from h1 to h6 as largest to smallest <h1> </h1>
- **Tag for Line Break:** To split a sentence into multiple lines a tag is used.
- **span:** tag is used to write content but it takes space of only content.
- **div:** <div> tag is used to write content but it takes space of entire width.

TAGS USED IN TEXT

Tag-Pair	Task
p	Paragraph, sentence
b	To make characters bold
i	Text is shown in italics
em	When you need to emphasize a word but with italics.
strong	When emphasize a word but with bold
sup	Superscript, helpful in formula and footnotes
sub	Subscript, helpful in formula and footnotes
u	Underline a text
small	Smaller text size, like footnote

TAGS IN HTML CONT....

- **img:** `` is used to insert image, src short for source is the path where the file is located.
- **Bullets:** `<ul>` stands for unordered list is used to define bullets. Bullets can be of type circle, square or disc.
- **Numbering:** `<ol>` stands for ordered list is used to define numbering.
- **Links:** `<a>` tag also called anchor tag or hyperlink used to generate link. A link can be text and image. In the link tag we have an attribute – target which define whether the link would open in same or new tab.
- **Table :** `<table>` tag is used to generate table. There are other multiple tags used inside table tag which will discuss in practical task referring the textbook.
- **Video:** `<video>` tag is used to add video.

CASCADING STYLE SHEETS (CSS)

- The CSS is used to add designing in the HTML content. For instances; colors, layout, font style etc.
- In HTML, there are three ways to incorporate CSS; inline, internal, external.
- Inline CSS: in this type, we applying styling on tag in the opening tag.
- Internal CSS: in this type, we apply styling using style tag inside in the head.
- External CSS: in this type, we apply styling by integrating external CSS file link.

JAVASCRIPT (JS)

- JavaScript is a lightweight dynamic programming language that provides interactivity and logic into HTML content.
- Few common examples of JS applications are:
 - *Pop-up boxes, drop-down menus, image carousels, and search bar suggestions.*
 - *Fetching and loading new information in the background without forcing you to manually refresh the page (like a social media feed).*
- JavaScript code can be embedded in HTML with starting and ending tag of `<script>`, in a webpage.
- There is no limitation of where to place the code inside a HTML file.

JAVASCRIPT (JS) CONT....

- **Variable:**

- *A variable is an entity that stores some value for later use.*
- *var, let keywords are used to declare variables in JS.*
- *Variable hold the data of numbers, string, Boolean value.*
- *JS is case sensitive; meaning 'reward' and 'Reward' are two different variables.*
- *Before using any variable, it must be declared otherwise JS will throw error.*

- **Operators:**

- *JS supports arithmetic operators.*
- *Modulus (%) operator is an additional operator used to find remainder.*

JAVASCRIPT (JS) CONT....

- **Conditional Statement:**

- *Conditional statement is a core feature of programming which help decide a choice based on some criteria.*
- *The keywords 'if', 'else if', 'else' are used to apply conditional statement.*

Operator	Name	Example
==	Is equal	x == y
!=	Is not equal	x != y
>	Greater than	x > y
<	Less than	x < y
>=	Greater than or equal to	x >= y
<=	Less than or equal to	x <= y

JAVASCRIPT (JS) CONT....

- **Iterative Statement:**

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- **Nested loops:**

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- **Arrays:**

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- **Functions:**

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